IIT Madras

DL & GenAI Project Report

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Music Genre Classification

Final Kaggle Score: 0.88 Macro F1

01 Abstract

This report covers the Messy Mashup Kaggle competition where the goal was to classify music into one of ten genres from noisy audio mashups. Four models were built and evaluated: LightGBM and XGBoost on handcrafted features as a classical ML baseline (Milestone 2), a CNN from scratch on Mel spectrograms (Milestone 3), a second CNN with improved cross-song mashup augmentation (Milestone 4), and finally a fine-tuned Audio Spectrogram Transformer (AST) which achieved 0.88 Macro F1 on Kaggle (Milestone 5). All experiments were tracked with Weights and Biases.

Metric	Value
Genres	10 (Blues to Rock)
Final Kaggle F1	0.88 (Macro F1)
Best Model	AST Transformer (fine-tuned)
Total Models Built	4 (LGB · CNN · CNN-Augmented · AST)
Training Samples	5,000+ synthetic mashups

02 Introduction

The competition provided individual stems (drums, vocals, bass, other) per song rather than pre-mixed tracks. The test set is made by mixing stems from different songs of the same genre with tempo changes and ESC-50 noise added on top. The model cannot rely on recognising a specific song — it must understand what makes a genre sound the way it does. The evaluation metric is Macro F1, which treats all 10 genres equally.

Project Objectives

- Build four progressively complex models and compare them at each milestone
- Use augmentation that closely matches how the test mashups are constructed
- Demonstrate why a pretrained transformer outperforms a model trained from scratch
- Log all experiments in W&B and achieve 0.85+ Macro F1 on the Kaggle leaderboard

03 Dataset and Preprocessing

Part	Files / Count	Description
genres_stems/	4,000 stems	10 genres x 100 songs x 4 stems
ESC-50	2,000 clips	Environmental noise (50 categories)
Mashups	3,020 files	Unlabelled test mashups

EDA Findings

- All 10 genres have exactly 100 songs each — perfectly balanced dataset
- Most audio files are 20-30 seconds long; classical looks very distinct on spectrograms
- ESC-50 covers rain, traffic, crowd, and mechanical sounds — good noise variety

Audio Preparation

- Loaded at 16,000 Hz (Milestone 3/5) or 22,050 Hz (Milestone 4), amplitude normalised by peak value, target length 20s
- Training: random crop each time (free augmentation). Validation: fixed centre crop

Augmentation Strategy

- Cross-song mixing: each stem from a DIFFERENT song in the same genre (used in Milestones 4 and 5)
- Tempo stretch (p=0.4, range 0.88-1.12), random gain (0.7-1.3), ESC-50 noise (p=0.7)
- SpecAugment: randomly zeros frequency bands and time frames on the spectrogram

04 Feature Extraction

AST Feature Extractor (Milestone 5)

ASTFeatureExtractor from HuggingFace converts raw audio into log-mel filterbank features using 128 Mel bins and 10ms hop length. Using the same extractor as pretraining is critical because the model expects a specific input format — it reads the 2D spectrogram as a sequence of small patches.

Mel Spectrogram for CNN Models (Milestones 3 and 4)

128 Mel bins, FFT size 2048, hop length 512 computed with librosa. Power converted to decibels then normalised to 0-1 range, giving a 2D image of shape (128, time) that convolutional layers process like a picture.

Handcrafted Features for Classical ML (Milestone 2)

Feature	Dims	What it Captures
MFCC mean + std	80	Timbre and texture of sound
Chroma mean + std	24	Harmonic and melodic content
Spectral contrast	7	Peaks vs valleys in frequency spectrum
Mel spectrogram mean	64	Overall energy across frequency bands
Zero crossing rate + RMS	2	Percussiveness and loudness

05 Models Built

INPUT: All CNN/AST models receive shape (B, 1, 128, T) — Batch x Channel x MelBins x Frames

Milestone 2 — Classical ML Baseline (LightGBM + XGBoost)

LightGBM used 500 trees with learning rate 0.05 and 63 leaves. XGBoost used 300 trees with learning rate 0.05 and max depth 6. Both used StandardScaler and a stratified 85/15 split. Features were 177-dimensional handcrafted audio features (MFCCs, chroma, spectral contrast, etc.). These establish the floor — showing how much better deep learning performs compared to manual feature engineering.

Milestone 3 — CNN from Scratch

Four convolutional blocks each containing Conv2d, BatchNorm2d, ReLU, MaxPool2d, and Dropout2d. Channel depth increases 32 → 64 → 128 → 256. Global Average Pooling produces a fixed-size vector which is fed to a two-layer fully connected classifier. Trained with AdamW optimizer (lr=3e-4), cosine LR schedule, label smoothing (0.1), and SpecAugment. Dataset used genre-level stems mixed per song.

Milestone 4 — Enhanced CNN with Cross-Song Mashup Augmentation

Uses the same CNN architecture (4 convolutional blocks, 32 → 64 → 128 → 256 channels, Global Average Pooling) as Milestone 3, but introduces a significantly improved data loading strategy. The CNNMashupDataset creates synthetic training samples by mixing stems from DIFFERENT songs of the same genre (cross-song mixing), closely mimicking how the actual test mashups are constructed. Additional augmentations include random gain (0.5-1.5), ESC-50 noise injection, and SpecAugment. The improved augmentation pipeline is the key differentiator over Milestone 3.

Milestone 5 — Fine-Tuned AST Transformer

The Audio Spectrogram Transformer (AST) divides the spectrogram into 16x16 patches, embeds them, adds positional encodings and a CLS token, then processes the sequence through 12 transformer encoder layers. Pretrained on AudioSet (527 classes, 2M clips). Final classifier replaced with a 10-class head.

Two-phase training strategy:

- Phase 1 (3 epochs): Base model frozen at LR=3e-4. Only the classifier head is trained — prevents damaging pretrained AudioSet weights.
- Phase 2 (5 epochs): All layers unfrozen. Base LR=2e-5 (very small to protect pretrained weights), classifier LR=2e-4 (10x larger as head still needs adaptation). Gradient accumulation (steps=2) gives effective batch=12. Cosine warmup scheduler for smooth convergence.

06 Results and Analysis

Final Model Comparison

Model	Val F1	Kaggle F1	Notes
LightGBM	~0.45	0.40	177-dim handcrafted features
XGBoost	~0.48	0.43	Same features, different booster
CNN from Scratch (M3)	~0.55	0.48	Mel spectrogram + SpecAugment
Enhanced CNN with Mashup Aug (M4)	~0.60	0.55	Cross-song mixing + noise injection
AST Phase 1	0.83	0.82	Frozen base, classifier head only
AST Phase 2	0.8596	0.88	Full fine-tune — BEST MODEL

Key Observations

- Each milestone shows a clear step-up: 0.45 (LGB) → 0.55 (CNN) → 0.60 (CNN+Aug) → 0.88 (AST)
- AST wins because it was pretrained on 2 million AudioSet clips before fine-tuning on only 1,000 songs
- Cross-song mashup augmentation (M4) improved the CNN significantly by making training data match test conditions
- Two-phase training was critical — freezing the base first prevented catastrophic forgetting of AudioSet knowledge
- Test Time Augmentation (5 crops averaged) pushed val F1 from 0.8596 to Kaggle 0.88

Error Analysis

- Disco and hiphop confused most often — both use electronic beats and similar rhythms
- Country and blues sometimes mixed — both rely heavily on guitar with similar timbre
- Classical music most reliably classified — sparse harmonic structure is very distinct
- Reggae occasionally confused with pop — both melodic, though reggae has a distinctive off-beat

07 Conclusion

Main Takeaways

- Transfer learning from AudioSet is extremely powerful. Fine-tuning on 1,000 songs is enough to reach 0.88 F1 because the model already understands audio deeply.
- Matching augmentation to test conditions is key. Cross-song mixing, tempo changes, and noise in training (introduced in Milestone 4) directly mirrors how test mashups are built — this alone gave a measurable jump over the basic CNN.

- Two-phase training protects pretrained weights and then adapts them smoothly with layer-wise learning rates.
- Improved data augmentation (M4 CNN) outperformed the simpler augmentation CNN (M3) even with the same model architecture, confirming that data quality matters as much as model complexity.

Challenges Along the Way

- Kaggle session expired before model weights were saved — full retrain required
- Stem filename was other.wav not others.wav, causing all songs to be silently skipped until found
- Gradient accumulation was in config but missing from the training loop until caught and fixed

What Could Be Done Next

- Ensemble three AST models with different seeds by averaging probabilities — could reach 0.93-0.95
- Longer audio (30 seconds), more synthetic samples (10,000+), and W&B hyperparameter sweeps
- Try HuBERT or Wav2Vec2 as alternative pretrained models for complementary ensembling
- Deploy as a Gradio app on HuggingFace Spaces for live genre prediction from uploaded audio

08 Experiment Tracking — Weights and Biases

All milestones were tracked in the W&B project 23f3000162-t12026. Each model run logged train loss, train F1, and val F1 after every epoch. The dashboard allowed fair comparison across all experiments without manual record-keeping.

W&B Run Name	Phase	Best Val F1
milestone1-eda	EDA and baseline analysis	—
milestone2-lightgbm	Classical ML training	~0.45
milestone2-xgboost	Classical ML training	~0.48
milestone3-cnn	CNN from scratch	~0.55
milestone3-cnn (M4 run)	Enhanced CNN with mashup augmentation	~0.60
milestone5-ast-phase1	Classifier warmup (frozen base)	~0.83
milestone5-ast-phase2	Full fine-tune (all layers)	0.8596

Key Hyperparameters — AST Phase 2 (Best Run)

Parameter	Value	Reason
Sample Rate	16,000 Hz	Standard for audio AI models
Audio Duration	20 seconds	Covers full rhythmic and melodic patterns
Batch Size	6	Small due to large model (87M params) on GPU
Gradient Accumulation	2	Effective batch = 12 without extra GPU memory
Phase 1 LR	3e-4	High LR safe when base is frozen
Phase 2 Base LR	2e-5	Very small to protect pretrained weights
Phase 2 Head LR	2e-4	10x bigger — head still needs more adjustment
TTA crops	5	Averaged predictions for higher inference accuracy
Synthetic train size	5,000	Generated from 850 real song folders

Training Infrastructure

- Platform: Kaggle Notebooks with GPU T4 x2 accelerator
- Total training time: approximately 2.5 to 3 hours for AST Phase 1 and Phase 2 combined
- Model size: 87 million parameters, saved checkpoint 328 MB
- Libraries: PyTorch, HuggingFace Transformers, librosa, torchmetrics, wandb

09 References

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